GenAI Unit 3 Lab – Assignment

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Section	C

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your sessio
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% 794/794 [00:00<00:00, 29.3kB/s]
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 457MB/s]
Loading weights: 100% 105/105 [00:00<00:00, 403.50it/s, Materializing param=classifier.weight]
BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniLM-L-6-v2
Key                          | Status     |  |
-----------------------------+------------+--+-
bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 1.33k/? [00:00<00:00, 46.8kB/s]
vocab.txt: 232k/? [00:00<00:00, 7.32MB/s]
tokenizer.json: 711k/? [00:00<00:00, 28.4MB/s]
special_tokens_map.json: 100% 132/132 [00:00<00:00, 9.80kB/s]
README.md: 3.67k/? [00:00<00:00, 166kB/s]
```

```
modules.json: 100% 349/349 [00:00<00:00, 29.0kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 9.00kB/s]
README.md: 10.5k/? [00:00<00:00, 991kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.08kB/s]
config.json: 100% 612/612 [00:00<00:00, 60.4kB/s]
model.safetensors: 100% 90.9M/90.9M [00:00<00:00, 210MB/s]
Loading weights: 100% 103/103 [00:00<00:00, 504.85it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L-6-v2
Key                      | Status     |  |
-------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 13.3kB/s]
vocab.txt: 232k/? [00:00<00:00, 6.85MB/s]
tokenizer.json: 466k/? [00:00<00:00, 15.0MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 9.16kB/s]
config.json: 100% 190/190 [00:00<00:00, 17.0kB/s]
```

```
Loading weights: 100% 103/103 [00:00<00:00, 470.75it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L-6-v2
Key                      | Status     |  |
-------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
Query: how do transformers encode meaning?
Naive Top Doc: Transformers use self-attention to process input tokens in parallel.
Query: how do transformers encode meaning?

[HyDE]: Transformers, a type of neural network architecture, encode meaning by leveraging the power of self- ...

[Retrieved Docs]:
- Transformers use self-attention to process input tokens in parallel.
- Attention allows models to focus on important words in a sequence.
- Gradient descent is used to optimize neural networks.
- The BLEU score is a metric used to evaluate machine translation output quality.
- Convolutional networks are used for image processing tasks.

[Top Docs After Rerank]:
- Transformers use self-attention to process input tokens in parallel.
- The BLEU score is a metric used to evaluate machine translation output quality.
- Convolutional networks are used for image processing tasks.

[Final Answer]:
 Transformers use self-attention to process input tokens in parallel.
Query: optimization techniques for training
Naive Top Doc: Gradient descent is used to optimize neural networks.
Query: optimization techniques for training

[HyDE]: Optimization techniques play a crucial role in training machine learning models, as they enable the  ...

[Retrieved Docs]:
- Gradient descent is used to optimize neural networks.
- Adam optimizer improves convergence using momentum.
- Attention allows models to focus on important words in a sequence.
- Convolutional networks are used for image processing tasks.
- The BLEU score is a metric used to evaluate machine translation output quality.

[Top Docs After Rerank]:
- Gradient descent is used to optimize neural networks.
- Adam optimizer improves convergence using momentum.
- Convolutional networks are used for image processing tasks.

[Final Answer]:
 Gradient descent and Adam optimizer.
Query: what is attention?
Naive Top Doc: Attention allows models to focus on important words in a sequence.
Query: what is attention?
```

```
[HyDE]: Attention is a complex cognitive process that enables individuals to selectively focus on specific s ...

[Retrieved Docs]:
- Attention allows models to focus on important words in a sequence.
- Gradient descent is used to optimize neural networks.
- Transformers use self-attention to process input tokens in parallel.
- Convolutional networks are used for image processing tasks.
- Recurrent networks process sequential data step by step.

[Top Docs After Rerank]:
- Attention allows models to focus on important words in a sequence.
- Transformers use self-attention to process input tokens in parallel.
- Gradient descent is used to optimize neural networks.

[Final Answer]:
 Attention allows models to focus on important words in a sequence.
```

Bonus 1:

```
Loading weights: 100%|▰▰▰
```

```
[50-word chunks] Top doc:
  and TPUs. The core innovation of the Transformer is the multi-head self-attention layer. In this layer, each token attends to every other token in the sequence, computing a weighted sum of value vectors guided by query-ke
Loading weights: 100%|▰▰▰
```